

Jonathan Pereira Miller

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<https://www.instagram.com/snailresearcher/>

Education

- Oct 2015 – May 2021* **Autonomous University of Madrid, Spain**
PhD Biology. Phylogeny, taxonomy and distribution of the family Hydrobiidae
Madrid, Spain
- Sep 2014 – Jul 2017* **Autonomous University of Madrid, Spain**
Master in Ecology
Madrid, Spain
- Sep 2010 – Jun 2012* **University of Havana, Cuba**
Master in Animal Ecology and Zoology
Havana, Cuba
- Sep 2003 – Jul 2008* **University of Ciego de Ávila, Cuba**
Agricultural Engineering
Ciego de Ávila, Cuba

Complementary training

- 2019 Systematics, Morphology and Evolution of Marine Molluscs. National Scientific Centre of Marine Biology, Russian Academy of Sciences. ForBio and MEDUSA course. Imparted by Dr Yuri Kantor, Dr Elena Krylova, Dr Manuel Malaquias, Dr Christiane Todt, Prof Louise Allcock, Dr Anton Chichvarkhin, Dr Gennady Kamenev, Dr Elena Vortsepneva and Dr Irina Ekimova.
- 2019 II Workshop GBIF.ES: Elysia, an application for the management of collections of natural history (GBIF)
- 2018 Theoretical-Practical course on Zoological Nomenclature (MNCN – CSIC). Imparted by Dr Miguel Ángel Alonso Zarazaga. 20h
- 2018 Advanced regression techniques in natural sciences and their application with R (MNCN – CSIC). Imparted by Dr Luis M. Carrascal. 21h
- 2017 VIII Online workshop on data tabulation for natural history collections technicians. GBIF Spain Coordination Unit. 30h
- 2017 Course “Management and Conservation of Collections of Natural History” organized by Universidad Complutense de Madrid. 60h
- 2016 Management methods to analyse multivariate data in ecology and conservation biology (MNCN – CSIC). Imparted by Dr Salvador Herrando Perez. 40h

- 2016 Advances and Quantitative Methods in Biogeography and Ecology I: Exploratory Analysis of Data and Linear Models in R
Universidad de Alcalá de Henares, Madrid, Spain. 24h
- 2016 Advances and Quantitative Methods in Biogeography and Ecology II: Islands Biogeography
Universidad de Alcalá de Henares, Madrid, Spain. 24h
- 2016 Advances and Quantitative Methods in Biogeography and Ecology III: Phylogenetic and Functional Diversity
Universidad de Alcalá de Henares, Madrid, Spain. 24h
- 2014 SEAG0211/MF1971_3. Regulations and internal policy of environmental management in the organization Imparted by the government of the autonomous community of Madrid.
- 2014 SEAG0211/MF1974_3 Prevention of Environmental Risk. Formative unit of the Environmental Management course. Imparted by the government of the autonomous community of Madrid.
- 2013 Graphic Design, Discovery Academy, Madrid. 60h
- 2013 Web Design, Discovery Academy, Madrid. 54h
- 2012 Analysis of Ecological Data, University of Havana, Cuba. 60h
- 2012 Advanced Biostatistics for Ecologists, University of Havana, Cuba.
- 2011 Work methodology in Ecology, University of Havana, Cuba. 60h
- 2011 Reading, Writing and Communication of Sciences, University of Havana, Cuba. 45h
- 2011 Principles of Zoological Taxonomy, University of Havana, Cuba. 75h
- 2011 Biodiversity, University of Havana, Cuba. 20h
- 2011 Research Methodology, University of Havana, Cuba. 90h
- 2011 Geometric Morphometry, University of Havana, Cuba
- 2010 Elements of Stratigraphy and sedimentology, 2010, National Museum of Natural History, Havana, Cuba. 20h
- 2010 General Taphonomy and Cuba Fossil Record, National Museum of Natural History, Havana, Cuba. 20h

Research/Work Experience

- Sep 2020 – Present **Postdoctoral Research Contract**
National Museum of Natural History, Spain.
Working on systematic of family Hydrobiidae (Mollusca: Caenogastropoda) with other poorly studied genera under the supervision of Dr Marian A. Ramos.
- Jan 2016 – Jan 2020 **PhD Student**
National Museum of Natural History, Spain.
Finishing my PhD thesis entitled “*Systematic, biogeography and evolutionary patterns of family Hydrobiidae (Mollusca: Caenogastropoda)*” under the supervision of Dr Marian A. Ramos and Dr Diana Delicado
- Sep 2017 – Dic 2017 **Research Internship (associated with the FPI fellowship)**
Training in Molecular Biology, Biogeography and Data Analysis
Supervised by Dr Tom Wilke and Dr Diana Delicado, Justus-Liebig-Universität Gießen, Germany.
- Jun 2017 **Master's Student**
Master dissertation: “*Effects of climatic, geographic and evolutionary variables on the global species richness distribution of the family Hydrobiidae s.str. Stimpson, 1865 (Caenogastropoda)*”. Supervised by Dr Enrique García-Barros, Autonomous University of Madrid, Spain
- Oct 2014 – Jan 2017 **Master's Student**
Universidad Autónoma de Madrid, Department of Ecology

Madrid, Spain.

Diet analysis of Dupont's Lark (Chersophilus duponti). Training on insect taxonomy. Supervised by Dr Juan Traba, Autonomous University of Madrid, Spain

- Jul – Aug 2014 **Professional Training** in North Portugal, on the project “*Working on the endangered freshwater mussel Margaritifera margaritifera, on ecological niche and environment*”, fieldwork supervised by Dr Joaquim Reiss, University of Lisbon, Portugal.
- Feb 2011 – Jun 2012 **Master's Student**
Geometric Morphometrics on land snails of Cuba, genus Cerion.
Supervised by Dr Denis Dennis, University of Havana, Biology Faculty Havana, Cuba.
- Sep 2007 – Jun 2008 **Final degree project for Agricultural Engineering**
“*Structural characterization and primary productivity of red mangrove (Rhizophora mangle), Cayo Coco, Cuba*” supervised by MSc. Raul Gomez, University of Ciego de Ávila, Cuba.

Awards & Grants

- Jan 2016 – Jan 2020 FPI contract BES-2015-073953 linked to the Fauna Ibérica XI CGL2014-53332-C5-1-P project and Fauna Ibérica XII PGC2018-095851-B-C61, for a PhD research on “Systematic, taxonomy, biogeography and Phylogenetic on the family Hydrobiidae” at the National Museum of Natural Sciences, Spain.
- Sep 2014 – Dic 2015 Work fellowship on the International Relationship Department at the Faculty of Sciences (Universidad Autónoma de Madrid), managing the mobility programs for European students.
- 2014 Collaboration Grant (Ministry of Education, Spain) at the Department of Ecology (Universidad Autónoma de Madrid): Diet analysis of Dupont's Lark (*Chersophilus duponti*). Training on insect taxonomy.
- 2011 Award: Premio Gustavo Furrázola. Cuban Geology Society

Skills & Activities

- Skills** Biodiversity, Ecology and Evolution, Species Distribution Models, Taxonomy, Systematics, Invasive Species, Malacology, Speleology, Land Snails Ecology, Evolution, Phylogenetic Analysis, Conservation Biology, Geometric Morphometrics, Molecular Phylogenetic, Bioinformatics (R programming), Paleontology, Paleomalacology, Micro CT scanning, ESEM, Histology techniques.
- Languages** English, French, German, Italian, Russian
- Scientific Memberships** Spanish Society of Malacology, Cuban Society of Speleology
- Research Interests** Taxonomy, Zoology, Malacology, Paleomalacology Ecology, Modelling, Bioinformatics, Systematic, Phylogeny.

Publications

- Miller, J. P.**, Delicado, D., García-Guerrero, F., Ramos M. A. 2022. Recurrent founder-event speciation across the Mediterranean likely shaped the species diversity and geographic distribution of the freshwater snail genus *Mercuria* Boeters, 1971 (Caenogastropoda: Hydrobiidae). *Molecular Phylogenetics and Evolution*. In press
- Martínez, J., Carrillo, M., **Miller, J.P.**, 2022. First record of a rare sinistral specimen of *Pyramidula jaenensis* (Clessin, 1882) (Gastropoda: Pyramidulidae) in Spain. *Folia Malacologica*. <https://doi.org/10.12657/folmal.030.009>
- Miller, J.P.**, García-Guerrero, F., Rodríguez Sousa, A.A., Carrillo Pacheco, M. 2021. Taxonomic redefining of *Pyramidula jaenensis* (Clessin, 1882) (Gastropoda, Pyramidulidae) based on an integrative taxonomy approach. *Invertebrate Zoology*, 18(4): 465–480.
- Miller, J. P.** 2021. Systematics, biogeography and evolutionary patterns of the Hydrobiidae family (Mollusca: Caenogastropoda). Autonomous University of Madrid., Madrid. pp. 374.
- Miller, J.P.** 2021. On the validity of the name *Pseudamnicola subproductus* (Paladilhe, 1869) (Caenogastropoda, Hydrobiidae). *Folia Malacologica* 29(4): 234–237. <https://doi.org/10.12657/folmal.029.022>
- Arconada, B., **Miller, J. P.**, García Guerrero, F., Ramos M. A. 2019. Pasando desapercibidos, la gran diversidad de unos diminutos caracoles, los hidróbidos. *Naturalmente* 24: 91–96
- Rodríguez Sousa, A. A., et al. 2019. Physiological study of pulmonary involvement in adults with cystic fibrosis through simulated modelling of different clinical scenarios. *Medical & Biological Engineering & Computing* 57(2): 413–425. <https://www.ncbi.nlm.nih.gov/pubmed/30171436>.
- Miller, J. P.**, Ramos, M. A., Hauffe, T., & Delicado, D. 2018. Global species richness of hydrobiid snails determined by climate and evolutionary history. *Freshwater Biology*, 63(10): 1225–1239. doi:10.1111/fwb.13128.
- Miller J. P.** 2016. Geometric morphometric analysis of the shell of *Cerion mumia* (Pulmonata: Cerionidae) and related species. *Folia. Malacologica* 24: 239–250. <http://dx.doi.org/10.12657/folmal.024.020>.
- Pereira-Miller F.J.** 2015. Inventario de los moluscos terrestres de Boquerones, Ciego de Ávila, Cuba. *Revista peruana de biología* 22(2): 239–245. <http://dx.doi.org/10.15381/rpb.v22i2.11358>.

Conference Proceedings

- García-Guerrero F.; **Miller J.P.**; Ramos M.A., 2021. Unravelling the taxonomy of the genus *Belgrandia* Bourguignat, 1870 (Mollusca: Hydrobiidae) in the Iberian Peninsula through the use of multilocus species delimitation methods. V PhDay Biology, Universidad Complutense de Madrid, Spain.
- García-Guerrero F.; **Miller J.P.**; Delicado D.; Ramos M.A., 2021. Morphological species of the genus *Belgrandia* Bourguignat, 1870 (Mollusca: Caenogastropoda: Hydrobiidae) in the Iberian Peninsula evaluated using multilocus species delimitation methods. 9th European Congress of Malacological Societies, Prague.
- Miller, J P.**; Delicado, D.; Ramos, M A., 2019. *Disentangling taxonomy of Mercuria Boeters, 1971 using an integrative approach*. International Seminar on Biodiversity and Evolution of Mollusks, September 26–27, 2019, Vladivostok, Russia. NSCMB FEB RAS, p. 47
- Miller, J P.**; Ramos, M A.; Hauffe T.; Delicado, D., 2017. *Effects of climatic, geographic and evolutionary variables on the global species richness distribution of the family Hydrobiidae s.str.*

Stimpson, 1865 (Caenogastropoda). 8th EUROMAL Congress 2017, Krakow, Poland. ISBN 978-83-7986-156-9, p. 92

Aguado, A.; **Miller, J P.**; García-Guerrero, F.; Delicado, D. y Ramos, M. A. *Diversidad, procesos evolutivos y biogeografía de especies de Hydrobiidae (Mollusca, Caenogastropoda) ibéricas*. I Jornadas de Investigación del MNCN, Spain. 02/2017

Miller, J P.: *Land snail biodiversity from a protected area in central Cuba. Threats and Future*. TiBE – Tropical Biodiversity. Oporto, Portugal. 11/2016

Miller, J.: *Sistemática, biogeografía y patrones evolutivos de la Familia Hydrobiidae (Mollusca: Caenogastropoda)*. III Congreso de Novedades en Investigaciones Zoológicas. Universidad Complutense de Madrid, Spain. 03/2016

Miller, J P.: *Análisis morfométrico de Cerion sp. (Mollusca: Cerionidae) procedente de dos localidades del Pleistoceno de la Costa Norte de la Habana, Cuba*. VIII Jornadas del departamento de biodiversidad y biología evolutiva - Museo Nacional De Ciencias Naturales - CSIC, Madrid; 11/2015

Miller, J.: *Catalog of fossil vertebrate localities of the quaternary in the province of Ciego de Ávila, Cuba*. Proceedings of the 7th European on REgional GEOscientific Cartography and Information Systems, Bolonia, Italy; 06/2011

Pereira-Miller, J., Gómez D., Pérez Y.: *Reporte de una nueva localidad para los peces del género Lucifuga (Ophidiiformes: Bythitidae) en la provincia de Ciego de Ávila, Cuba - Congreso 70 Aniversario Sociedad Espeleológica de Cuba, Matanzas, 2010*. Congreso 70 Aniversario Sociedad Espeleológica de Cuba, Matanzas, Cuba; 08/2010

Pereira-Miller, F J.: (2010). *Los moluscos terrestres de la Sierra de Jatibonico, Municipio Florencia, provincia Ciego de Ávila, Cuba*. Congreso 70 Aniversario de la Sociedad Espeleológica de Cuba Matanzas, Cuba.

Pereira-Miller, F J.: (2005). *Hallazgo paleontológico de relevancia en carso vertical de la provincia de Ciego de Ávila, Cuba*. Congreso 65 Aniversario de la Sociedad Espeleológica de Cuba Viñales, Cuba.

Pereira-Miller, F J.: (2005). *Las Trampas Térmicas en la provincia de Ciego de Ávila, Cuba*. Congreso 65 Aniversario de la Sociedad Espeleológica de Cuba Viñales, Cuba.

Pereira-Miller, F. J. (2005). *Breve bosquejo sobre la quiropteroфаuna local de la provincia de Ciego de Ávila, Cuba*. Congreso 65 Aniversario de la Sociedad Espeleológica de Cuba Viñales, Cuba.

Teaching Experience and supervising

January 2022. Analysis of the diversity of the genus *Bythinella* in the Iberian Peninsula. Supervisor to the Master Final Project in The Complutense University of Madrid.

September 2008 – July 2009. Lecturer in Organic Chemistry at the University of Ciego de Ávila, Cuba.

2005 – 2011 Teacher in several non-official short courses within the Cuban Speleological Society

- Course of Speleology: basic level
- Course of Speleology: medium level
- Course of Geology: basic level
- Course of Paleontology: basic level
- Course of Design and elaboration of research projects

Peer Review Experience

- 2018 Freshwater Biology Journal. Manuscript entitled: “*Modelling species distributions to predict areas at risk of invasion by the exotic aquatic New Zealand mud snail Potamopyrgus antipodarum (Gray 1843)*”
- 2019 European Journal of Taxonomy. Manuscript entitled: “*Two new genera and three new subterranean species of Hydrobiidae (Caenogastropoda: Truncatelloidea) from Tunisia*”
- 2020 Progress in Physical Geography. Manuscript entitled: “*The relevance of ecoregions and mountainous environments in the diversity and endemism of land gastropods*”
- 2021 Asian Journal of Research in Zoology. Manuscript entitled: “*Checklist of Gastropod and Bivalve Mollusca Recorded from Selected Sites of Pirojpur and Barguna District of Bangladesh*”
- 2021 Spira. Manuscript entitled: “*Diversity and distribution of stygobitic and stygophilous molluscs (Moitessieriidae and Hydrobiidae) in the Sant Llorenç del Munt i la Serra de l’Obac massif (Barcelona, NE Iberian Peninsula).*”
- 2022 Spira. Manuscript entitled: “*Greece, a second hotspot of Bythinella spp. (Gastropoda: Bythinellidae) in Europe, with the description of two new species*”